

AI Prompt Logs

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Project Name	AdForge AI: AI-Powered Advertisement Copy Generator
Maximum Marks	3 Marks

AI Prompt Logs:

The AI Prompt Logs document the structured interactions with Large Language Models during the development of AdForge AI. These logs serve as a transparent record of how artificial intelligence was leveraged to accelerate the development process, improve code quality, and solve complex technical and design challenges throughout the project lifecycle.

During the development of AdForge AI, our team used AI assistance across eight major development tasks including Flask backend architecture, prompt engineering, frontend UI design, JavaScript controller development, CSS styling, input validation, content safety filtering, and performance tips generation. Each prompt was carefully crafted with a specific system context that instructed the AI to act as a domain expert relevant to the task — such as a Senior Python Developer for backend tasks, a Senior Copywriter for marketing prompt engineering, and a Senior UI/UX Designer for frontend styling.

The prompt engineering process was iterative in nature. Initial prompts often produced functional but incomplete results that required refinement. For example, the Google Ads headline prompt required additional instructions to enforce the 30 character limit strictly and eliminate repeated headlines. The Groq API integration required version pinning of httpx to resolve a proxies keyword argument conflict. The CSS stylesheet required additional refinement to include glass morphism effects, markdown body table styles, and mobile responsive breakpoints.

These logs demonstrate that AI was not used as a replacement for developer thinking but rather as an intelligent accelerator that reduced development time significantly. Every AI output was reviewed, tested, refined, and integrated by the development team to ensure it met the project requirements, security standards, and quality expectations of AdForge AI.

The structured prompt engineering approach documented in these logs ensures full reproducibility of the development process and provides clear evidence of how generative AI tools were used responsibly and effectively to build a production-ready web application.

S · N o	Task / Intent	System Prompt & Context	Exact User Prompt	AI Output Summary	Refinemen ts / Iterations
1	<i>Generate Flask project structure and main app.py with Groq API integration</i>	<i>Act as an elite Full-Stack Python developer. Use Flask 3.0, Groq API, python-dotenv, and bleach. Follow MVC architecture and secure coding practices</i>	"Create a production-ready Flask app.py for an AI advertisement generator. Include Groq API client initialization, input sanitization using bleach, content safety filter with banned keywords, and three API endpoints: /api/generate, /api/refine, and /api/tips. Use environment variables for API key management"	<i>Generated complete app.py with Groq client setup, three modular route handlers, bleach sanitization, banned keyword safety filter, validate_inputs function, and proper error handling with HTTP status codes</i>	<i>Needed to prompt again to fix the proxies keyword argument error in Groq SDK. Added httpx==0.27.0 version pinning to resolve the conflict</i>
2	Build advanced marketing prompt engineering system	Act as a Senior Copywriter and Digital Marketing Strategist with 20 years of experience. You understand	"Build a prompt builder function in Python that takes campaign brief data including product name,	Generated build_ad_prompt() function with conditional platform sections, structured campaign brief table, specific character limit instructions for Google	Refined the prompt to add explicit instructions for 15 unique headlines with no repetition.

		AIDA, PAS, FAB, FOMO, Scarcity, and Social Proof frameworks	description, target audience, tone, USP, and selected platforms and constructs a structured marketing prompt that generates platform-specific Google Ads headlines under 30 characters, Instagram captions with hashtags, and Facebook primary text variations"	Ads, emoji and hashtag instructions for Instagram, and conversational tone instructions for Facebook	Added marketing psychology framework instructions including AIDA, PAS, and FOMO to improve output quality
3	Create Advertisement Refinement Engine prompt	Act as a Conversion Rate Optimisation Expert and Senior Copywriter. Analyse existing advertisements and improve them based on specific user feedback	"Build a Python function that takes an original advertisement and user feedback such as make it shorter, add urgency, or target teenagers and constructs a prompt that returns an improved version with a list of changes made, an improvement score out of	Generated build_refine_prompt() function that structures the original ad and feedback into a clear prompt. Output includes Improved Version, Changes Made as bullet list, Improvement Score with justification, and Conversion Prediction sections in Markdown format	Needed to add explicit Markdown section headings to ensure consistent output formatting. Added reasoning instruction telling the AI to analyse before rewriting for better quality results

			10, and a conversion prediction comparing original and improved versions"		
4	Design premium SaaS frontend UI with HTML and CSS	Act as a Senior UI/UX Designer specialising in premium SaaS products. Use HTML5, CSS3, and Vanilla JavaScript. Dark theme with violet and mint color palette	"Create a premium dark-themed single page web application for AdForge AI with a sticky navigation header, animated hero section with gradient text, three tab panels for Ad Generator, Refinement Engine, and Performance Tips, platform selection chips, a 18-field campaign brief form, loading spinner with rotating messages, and output panel with copy and download buttons"	Generated complete index.html with full navigation, hero section with gradient animated text, stats row, three functional panels with tab switching, custom platform checkbox chips, all form fields with proper labels and placeholders, loader container, and output area with toolbar	Refined to add smooth scroll behavior for navigation links. Added rotating loader messages in JavaScript. Adjusted CSS color variables for better contrast and accessibility on dark background
5	Build Performance Tips	Act as a Growth Marketing Expert and	"Create a Python prompt	Generated build_tips_prompt() function with	Refined to add platform-

	Generator prompt	Digital Advertising Specialist. Generate actionable, specific, and data-driven performance recommendations	builder function for a standalone Performance Tips Generator that takes product name, target audience, campaign goal, tone, and selected platforms and generates 4 A/B testing ideas with hypothesis, 3 CTA improvement variants, 3 SEO keyword tips, 3 conversion optimisation tips, platform-specific best posting times, and 2 to 3 refined audience segments"	structured prompt sections for each performance category. Output includes A/B Testing Suggestions, CTA Improvements, SEO Suggestions, Conversion Optimisation Tips, Best Posting Times, and Audience Recommendations in clean Markdown format	specific posting time instructions since initial output was too generic. Added hypothesis format requirement to A/B testing ideas to make them more actionable and useful
6	Implement input validation and content safety filter	Act as a Senior Python Security Engineer. Focus on input validation, sanitization, and content moderation for a Flask web application	"Create a Python safety filter function that checks user inputs against a list of 20 plus banned keywords related to illegal products,	Generated is_unsafe_input() function with comprehensive banned keyword list, sanitize() function using bleach.clean() with tags=[] and strip=True parameter, and validate_inputs() function that checks required fields and	Needed to add platform whitelist validation to prevent injection through platform selection field. Added maximum length

			fraud, scams, adult content, hate speech, violence, and spam. Also create a sanitize function using bleach that strips all HTML tags and enforces maximum character limits on all form fields"	platform selection returning a list of error messages	parameters to sanitize function for each field type based on expected input size
7	Create responsive CSS stylesheet with premium design	Act as a Senior Frontend Designer specialising in SaaS products. Use CSS custom properties, modern flexbox and grid layouts, and smooth animations	"Create a complete CSS stylesheet for AdForge AI with a dark near-black background, electric violet and neon mint accent colors, Syne display font and DM Sans body font, CSS custom property design tokens, animated gradient text effect, platform chip toggle styles, form input focus states with glow effect, loading spinner animation, output panel	Generated complete style.css with 50 plus CSS custom property design tokens, animated gradient text with hue-shift keyframes, premium card styles with hover effects, custom select dropdown styling, platform chip active states, spinner and pulse animations, markdown body styles for output formatting, and responsive breakpoints for mobile and tablet	Refined to add backdrop blur effect on sticky header for modern glass morphism look. Adjusted markdown body styles to include table formatting, blockquote styles, and code block highlighting for better AI output readability

			slide-up animation, and fully responsive mobile layout"		
8	Build JavaScript frontend controller	Act as a Senior Frontend JavaScript Developer. Use Vanilla JavaScript with async/await, FormData API, and modern DOM manipulation	"Create a complete script.js file that handles tab switching between three panels, platform checkbox chip toggle with active state, form data collection from 18 fields, async fetch calls to /api/generate /api/refine and /api/tips endpoints, rotating loader messages during API calls, Markdown rendering of AI output, copy to clipboard functionality, download as TXT file, and client side form validation with error display"	Generated complete script.js with initTabs() for panel switching, initPlatformChecks() for chip styling, submitAdForm() submitRefineForm() and submitTipsForm() async handlers, getCheckedPlatforms() helper, showOutput() and showAlert() UI helpers, copyOutput() using clipboard API with fallback, downloadOutput() using Blob and URL.createObjectURL, and cycleLoaderText() for rotating messages	Refined to add auto-dismiss error banners when user starts typing in required fields. Added safety check to clear loader interval after 30 seconds maximum to prevent memory leaks if API call hangs